

Renormalization as Standing-Preserving Scale Transport

RG Flow, Universality, Fixed Points, and Effective Field Theory under Admissibility Constraints

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Abstract

Renormalization-group flow is often introduced as the scale dependence of couplings, fields, and effective parameters. This paper develops a complementary structural account. Building on the fixed-domain result that standing-bearing constants are not freely variable same-domain parameters, and on the subsequent classification theorem separating admissibility-fixed, quotient-dependent, effective, transport-defined, and presentation-dependent roles, it argues that RG flow is best understood as lawful scale transport preserving standing-bearing physical content across admissible effective descriptions. Running couplings, scheme dependence, regulators, thresholds, fixed points, universality classes, and effective field theories do not undermine constant fixity or parameter role classification. They exhibit the physical mechanisms by which role structure is transported across scale.

The central theorem states that when scale-indexed descriptions are connected by RG maps preserving overlap witness structure, factoring through standing quotients, and respecting effective envelopes, scale-dependent parameter change is admissible transport rather than unlicensed same-domain variation. The standing-bearing content is not raw numerical sameness of coordinates such as $g(\mu)$, but the induced quotient transport class through which observables, matching data, and universal quantities retain identity across scale. Universality classes are thereby classified as quotient-persistent structures, fixed points as quotient-stable RG loci, scheme choices as presentation-dependent skin when witnesses are preserved, EFTs as envelope-bounded admissible descriptions, and threshold matching as boundary-trace fixation for regime transition. The paper further distinguishes formal RG families from standing-bearing continuation: a list of scales, coupling values, schemes, cutoffs, or effective theories does not itself constitute RG transport unless a continuation locus, overlap witnesses, quotient identity, envelope discipline, and target-domain typing are supplied. The result does not modify RG equations, derive beta functions, deny scheme dependence, replace effective field theory, or assert a new dynamics. It supplies an admissibility-theoretic normal form for interpreting what RG flow preserves, what it changes, and when scale-dependent variation is physically licensed.

Orientation. The companion fixed-domain analysis established the comparison class: standing-bearing constants are not freely variable same-domain parameters. The companion classification theorem separated the roles of constant-like quantities. The present analysis applies those results to the central case in which scale-dependent parameter change is both unavoidable and physically successful: renormalization. The thesis is that RG flow is a paradigm case of admissible scale transport, and that a formal flow family becomes physical RG continuation only when its standing-bearing continuation structure is fixed.

Contents

1	Introduction: RG as the Natural Bridge	3
1.1	The bridge from constants to renormalization	4
1.2	Main claim	5
1.3	Structure of the argument	5
2	Scope and Non-Claims	5
2.1	Claims	5
2.2	Non-claims	6
2.3	Relation to ordinary RG practice	6
3	Contribution Relative to Standard RG Formalism	7
4	Anchor, Tensor, Skin, and the RG Role Normal Form	8
5	Five RG Roles and Their Interactions	9
6	Scale-Indexed Admissibility Domains	10
7	RG Continuation Loci and Target-Domain Retyping	11
7.1	Continuous families and replicated sectors	12
8	Raw Coupling Labels versus Standing-Bearing RG Content	13
9	Overlap-Level Coincidence and Non-Divergence	14
10	The RG Transport Proposition	14
11	Scheme Dependence as Presentation-Dependent Skin	15
11.1	Universal and non-universal beta-function data	15
11.2	Regulators and cutoffs	16
12	Effective Field Theory as Envelope-Bounded Admissibility	16
12.1	Wilson coefficients and low-energy constants	17
13	Universality Classes as Quotient Persistence	18
13.1	Universality and quotient convergence	18
14	Fixed Points as Transport-Stable Quotient Loci	19
14.1	UV and IR roles	19
15	Relevant, Irrelevant, and Marginal Operators as Role Diagnostics	20
16	Thresholds, Matching, and Boundary-Trace Fixation	21
16.1	Matching as a transport condition	22
17	Dimensional Transmutation and Scale Anchors	22
17.1	The QCD scale as example	22
18	AMetric Boundary and No Default RG Selector	22

19 Null-Regime Exclusion for RG Endpoints	23
20 The RG Standing Preservation Theorem	23
20.1 The theorem ladder	25
21 Error Taxonomy	25
21.1 Why these errors recur	26
22 Worked Example: Gauge Coupling under RG	26
22.1 Role decomposition	27
22.2 Trajectory, scheme, fixed point, and transmutation	27
22.3 Three comparisons	28
23 Objection Matrix	30
24 Conclusion	31
A Conceptual Dependencies	31
B Index of Formal Results	32
C Glossary of Role Terms	33
D Scheme Change and Invariant Extraction	34
E Universality as a Quotient Diagram	35

1 Introduction: RG as the Natural Bridge

Renormalization is among the clearest cases in physics where raw symbols and physical content must be distinguished. A coupling written as g is not a complete physical datum until a scale, scheme, normalization, domain of validity, and witness relation have been fixed. A regulator does not by itself constitute a witness-bearing physical entity. A cutoff can be a calculational device, an envelope marker, or a physical threshold depending on role. A low-energy constant may be empirically meaningful without being primitive. A fixed point may be a coordinate condition in one description and a universal structural locus in another. Ordinary RG practice already distinguishes physical identity across scale from raw equality of numbers.

The present analysis makes this implicit discipline explicit. It does not require treating AASC as a replacement for RG formalism; it requires only the operational assumption that physical quantities admit stable identification across admissible observational and theoretical transport. It interprets RG flow as standing-preserving scale transport: a lawful relation between scale-indexed descriptions that preserves empirical identification on overlap, factors through quotient structure, and respects the envelope within which an effective description is admissible. This interpretation is not an alternative to Wilsonian renormalization or perturbative RG. It is a structural account of why RG change differs from unlicensed variation of constants.

The issue is especially important in light of the preceding fixed-domain and classification results. The fixed-domain analysis argued that physical constants are not freely variable same-domain explananda once standing-preserving reference is imposed. The classification analysis then showed that constant-like quantities occupy different roles: some are admissibility-fixed anchors, some

quotient-dependent invariants, some effective envelope-bound quantities, some transport-defined data, and some presentation-dependent coordinates. RG flow is the natural test case: couplings run and effective parameters change with scale, yet the practice is physically successful.

The organizing thesis is:

RG flow is not unconstrained variation; it is lawful transport. It changes scale-attached coordinates while preserving, when successful, the standing quotient by which physical content is re-identified across effective descriptions.

This perspective clarifies the standing roles of the main RG objects. Running couplings are not primitive scalar identities that persist by numerical sameness; they are effective, transport-defined coordinates attached to an RG trajectory. Scheme dependence need not signal physical ambiguity when it remains presentation-dependent skin. Universality classes are quotient objects generated by preserved standing-bearing content under coarse-graining, rather than raw similarity among microscopic models. Fixed points are more than coordinate zeros of beta functions when they preserve the standing quotient and admit invariant overlap observables. EFT is not thereby epistemically deficient; it is an envelope-bounded admissible domain whose coefficients have standing through matching, witness preservation, and transport discipline.

A later continuation and embedding analysis sharpens this point without changing the RG target. Formal multiplicity of values, scales, schemes, cutoffs, or model labels does not inherit standing-bearing transport. Such multiplicity is either exploratory bookkeeping, lawful continuation inside an already fixed RG domain, or a newly typed target domain with its own witnesses, quotient identity, transport maps, and failure conditions. This distinction is used below only to harden the RG interpretation; it does not convert the paper into an ensemble-collapse or uniqueness argument.

1.1 The bridge from constants to renormalization

The fixed-domain analysis targeted a specific misclassification: treating a standing-bearing constant as if it were a freely variable same-domain coordinate before the domain preserving physical reference had been fixed. The classification analysis generalized this by distinguishing five roles. RG flow belongs naturally to that classification because a single RG calculation may involve all five roles:

- (1) The law-form and field content of a chosen comparison class function as anchors.
- (2) Observable equivalence on overlap supplies quotient-dependent identity.
- (3) Wilson coefficients and running couplings are effective quantities, meaningful within envelopes.
- (4) Beta functions, matching maps, and flow trajectories are transport-defined structures.
- (5) Scheme choices, regulator coordinates, field normalizations, and chart labels are presentation-dependent when witnesses are preserved.

The present analysis therefore treats RG not as an exception to fixed-domain discipline, but as a concrete physical exemplar of admissibility-preserving role discipline. Constants need not remain numerically unchanged to retain standing. What persists is the quotient-level identity relation by which physical content is compared across descriptions.

1.2 Main claim

The main claim is:

An RG flow is admissible as physical scale transport only when it preserves standing-bearing witness structure on overlap, factors through the relevant standing quotients, and remains within or lawfully transfers between effective envelopes. Under these conditions, scale-dependent parameter change is not same-domain free variation. It is standing-preserving quotient transport.

The claim is not a new beta function, regularization prescription, or UV completion criterion. It is an interpretive theorem about what is being preserved by RG practice and what is not. The raw values of $g(\mu)$, Wilson coefficients, counterterms, regulator coordinates, and scheme labels are not individually the invariant target. The invariant target is the admissible transport class of descriptions that preserves witness equivalence and quotient-level physical standing.

1.3 Structure of the argument

Sections 2–3 define the scope and specify what the admissibility account adds beyond standard RG formalism. Sections 4–6 define the RG role normal form and the scale-indexed admissibility domain. Sections 8–10 distinguish raw coupling labels from standing-bearing RG content and state the RG transport proposition. Sections 11–12 classify scheme dependence and EFTs. Sections 13–15 treat universality, fixed points, and operator relevance. Sections 16–18 address threshold matching, dimensional transmutation, and selector discipline. Section 19 gives the null-endpoint constraint. Section 20 proves the RG standing preservation theorem. Sections 21–22 provide an error taxonomy and a worked gauge-coupling example. Section 23 closes with objections and replies.

2 Scope and Non-Claims

The analysis is conservative about the formal physics. It does not alter the formal machinery of RG. It clarifies the admissibility role of that machinery.

2.1 Claims

The analysis claims:

- (C1) RG flow is a lawful scale-transport structure, not unsupported reassignment of constants.
- (C2) The invariant target of admissible RG is not raw numerical sameness of running parameters but preservation of standing-bearing witness structure through quotient transport.
- (C3) Scheme, regulator, and normalization dependence are presentation-dependent when all standing-bearing witness content is preserved.
- (C4) EFTs are envelope-bounded admissible descriptions rather than deficient approximations.
- (C5) Universality classes are quotient objects: invariant content preserved by admissible coarse-graining and scale transport.
- (C6) Fixed points are quotient-stable RG loci whose physical standing depends on preserved witnesses, quotient structure, and envelope role.

- (C7) Threshold matching functions as boundary-trace fixation for regime transition.
- (C8) No pre-admissible default scale, scheme, regulator, cutoff, fixed point, or endpoint selector is canonically fixed without admissibility structure.
- (C9) Formal families of scales, schemes, regulators, EFTs, or parameter values acquire standing-bearing RG authority only when they preserve a continuation locus or are explicitly retyped as target domains.

2.2 Non-claims

The analysis does not claim:

- (N1) a new formalism replacing Wilsonian or perturbative RG;
- (N2) new beta functions;
- (N3) numerical derivation of gauge couplings or Wilson coefficients;
- (N4) denial of scheme dependence;
- (N5) denial of regulators or cutoffs as legitimate calculational tools;
- (N6) rejection of EFT methodology;
- (N7) proof of UV completion;
- (N8) proof that all fixed points are physically realized;
- (N9) denial that distinct theories may have distinct RG structures;
- (N10) that practical parameter scans lack value;
- (N11) that explicitly governed parameter-family, scheme-family, or EFT-family analyses are prohibited.

The target is role misclassification: treating a coordinate attached to transport as an independently variable physical scalar; treating presentation dependence as tensor load; treating an EFT coefficient as a primitive constant outside its envelope; treating a fixed point as selected without witness transport; treating a pre-admissible parameter space as if it already carried a canonical physical measure; or treating formal RG-family multiplicity as if it already supplied a standing-bearing continuation relation.

2.3 Relation to ordinary RG practice

The present account is compatible with ordinary RG practice and explains why its standard distinctions matter. The dependence of a coupling on μ is meaningful only after one specifies how the coupling is defined. Scheme changes may alter coefficients or coordinate forms without changing observables. Matching across thresholds is required because a lower-energy description does not inherit standing by dropping degrees of freedom alone. Universality is powerful because many microscopic descriptions lose non-standing-bearing detail while preserving a quotient-level critical profile. These are not additions to RG practice; they are the admissibility constraints already presupposed by it. The present analysis concerns the admissibility role of RG-preserved structures rather than replacement of established field-theoretic machinery.

3 Contribution Relative to Standard RG Formalism

The formal machinery of renormalization is taken as given. Standard RG already supplies beta functions, coarse-graining procedures, running couplings, fixed points, universality classes, scheme transformations, and EFT matching. The present analysis does not duplicate that machinery. It asks a different question: what must be preserved for scale-dependent descriptions to remain descriptions of the same standing-bearing physical content?

The contribution is role-theoretic and admissibility-theoretic. Standard RG gives the transport equations; the present account classifies the standing roles of the structures transported by those equations and the conditions under which their transport preserves physical identification. It distinguishes raw coordinate motion from admissible scale transport, separates presentation-dependent scheme data from quotient-level invariant content, and states when an EFT continuation or threshold crossing preserves empirical standing rather than extending notation alone.

Table 1: Standard RG formalism and the added admissibility interpretation.

Standard RG supplies	This analysis adds
Beta functions and flow equations	A criterion for when flow is standing-preserving transport rather than unconstrained parameter reassignment.
Running couplings and Wilson coefficients	A role classification: effective, transport-defined, quotient-bearing, or presentation-dependent depending on envelope and witness structure.
Scheme transformations and regulator choices	A skin/tensor distinction separating coordinate presentation from standing-bearing invariant load.
Universality and critical behavior	A quotient interpretation of what many microscopic descriptions share under admissible coarse-graining.
Fixed points and scaling regimes	A quotient-stability interpretation tied to preserved standing quotients, invariant overlap observables, and admissible reconstruction.
EFT matching and decoupling	A boundary-trace criterion for inheriting standing across changes of envelope or degrees of freedom.

This distinction is essential. The claim is not that RG must be reconstructed as an AASC construction. The claim is that successful RG practice already presupposes standing-preserving transport: observable agreement on overlap, quotient-consistent re-identification of physical content, and envelope discipline for effective descriptions. The admissibility framework supplies a normal form for these presuppositions.

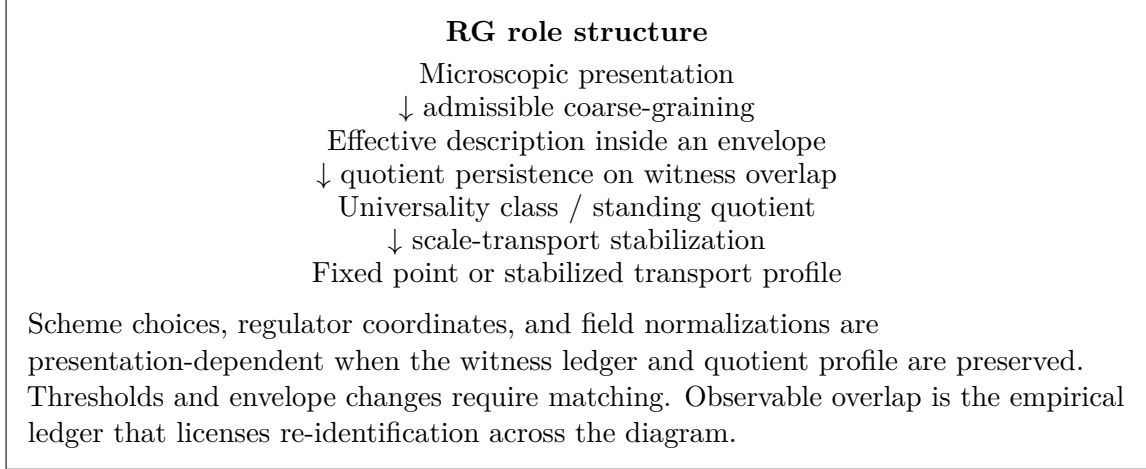


Figure 1: Role structure of RG transport. The diagram is not a new RG flow equation; it displays the admissibility roles that a successful RG relation must coordinate.

4 Anchor, Tensor, Skin, and the RG Role Normal Form

The RG setting is particularly well suited to the Anchor/Tensor/Skin distinction. Scale-dependent physics continually separates what fixes a domain, what carries invariant load, and what presents the calculation. The separation is not imposed as a metaphysical vocabulary. It formalizes distinctions already used in ordinary RG practice: a symmetry or field-content assumption is not a subtraction convention; a critical exponent is not a coordinate value; a cutoff does not by itself define a physical threshold; an EFT coefficient is not meaningful apart from its matching envelope.

Definition 4.1 (RG anchor/tensor/skin profile). *For an RG description, an anchor fixes the admissibility domain, law-form, symmetry class, field content, object-class, or envelope in which the flow is defined. A tensor carries standing-bearing invariant content under admissible RG redescription. A skin is a presentation-dependent coordinate, convention, scheme, regulator, normalization, or chart used to express the description.*

The same symbol may participate in more than one profile depending on the comparison class. A cutoff may be skin when it is a regulator to be removed, envelope when it marks the validity of an EFT, and boundary trace when it labels a matching surface. A coupling may be a transport coordinate at one level and a component of an invariant trajectory class at another. Role fixation precedes interpretation.

RG feature	Typical role	Interpretive discipline
Gauge group, symmetry class, field content	Anchor	Varying these usually changes the admissibility domain or object-class rather than a parameter inside it.
Critical exponents and universal scaling data	Tensor, when invariant under the admissible comparison class	These carry quotient-level physical load when connected to witnesses.
Raw coupling coordinate $g(\mu)$	Effective / transport-defined coordinate	Its standing comes through the RG trajectory and observables, not through raw numerical isolation.

Scheme, subtraction convention, field normalization	Skin, when witnesses are preserved	These organize presentation and can change coordinate values without changing standing.
Wilson coefficient	Effective	Physical within an EFT envelope and matching prescription.
Cutoff or regulator	Role-dependent	It may be skin, envelope, threshold, or physical structure depending on witness preservation.

Proposition 4.2 (ATS role discipline for RG). *An RG object is varied, interpreted, or promoted only relative to its anchor/tensor/skin profile. A skin-level change does not license tensor-level variation, and a domain anchor does not license treatment as a free same-domain coordinate.*

Proof. The proof is by role preservation. Skin changes preserve witness and quotient content by definition. Tensor changes alter standing-bearing content. Anchor changes alter the domain in which the content is defined. Treating one profile as another changes the comparison class and therefore blocks same-object interpretation. $\square$

This proposition is the operational bridge from the classification result to RG. It explains why RG can permit extensive coordinate motion without surrendering physical identity. The motion is admissible when it remains within the appropriate role profile and preserves the induced standing quotient.

5 Five RG Roles and Their Interactions

The five-role classification of constants becomes especially concrete under renormalization. RG practice uses all five roles at once, and confusions arise when they are collapsed.

- (R1) **Admissibility-fixed:** the law-form, symmetry class, field content, dimensionality, and allowed operator basis that define the RG problem.
- (R2) **Quotient-dependent:** universal long-distance content, critical data, equivalence classes of microscopic systems, and invariant relations extracted from RG histories.
- (R3) **Effective:** Wilson coefficients, running couplings at a scale, low-energy constants, and envelope-relative parameters.
- (R4) **Transport-defined:** beta-function flow, matching maps, scheme transformations, basis reconstruction, and dimensional-transmutation relations.
- (R5) **Presentation-dependent:** scheme labels, chart coordinates, regulator conventions, subtraction choices, and normalization conventions when witnesses are preserved.

Proposition 5.1 (RG role interaction). *A complete interpretation of an RG calculation requires all five roles. Removing one of them either destroys the comparison target, erases invariant physical load, loses envelope discipline, removes lawful transport, or mistakes presentation for physics.*

For example, a Wilson coefficient without an EFT envelope is not yet a physically interpretable coefficient. A beta function without a scheme and chart is not a complete coordinate expression. A fixed point without invariant scaling data is not yet a physical quotient-stable locus. A universality class without a quotient relation is only a list of systems. A regulator without role discipline invites unsupported cutoff ontology or premature dismissal.

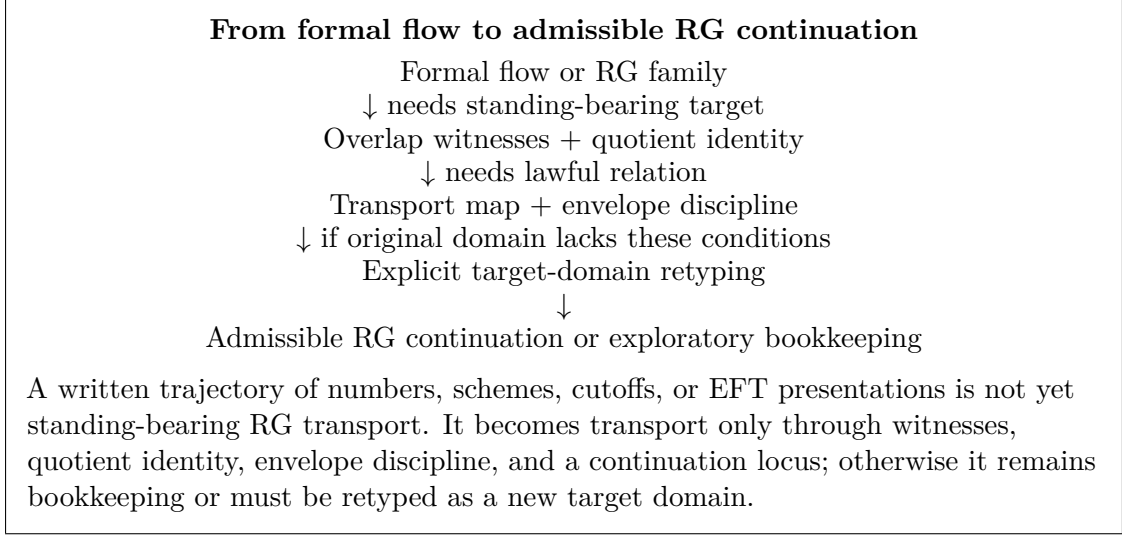


Figure 2: Theorem-flow diagram for RG continuation. The diagram separates formal scale-indexing from admissible scale transport.

6 Scale-Indexed Admissibility Domains

Definition 6.1 (Scale-indexed description). *A scale-indexed physical description is a structure*

$$D_\mu = (\mathcal{T}_\mu, \mathcal{A}_\mu, \mathcal{W}_\mu, \mathcal{R}_\mu, \mathcal{Q}_\mu, \mathcal{E}_\mu),$$

where $\mathcal{T}_\mu$ is the theory presentation at scale μ , $\mathcal{A}_\mu$ the admissibility conditions, $\mathcal{W}_\mu$ the witness ledger of observables and empirical identifications, $\mathcal{R}_\mu$ the lawful redescription structure, $\mathcal{Q}_\mu$ the standing quotient over admissibly equivalent presentations, and $\mathcal{E}_\mu$ the effective envelope specifying regime, cutoff, approximation, matching, and boundary data.

Definition 6.2 (RG transport map). *An RG transport map from scale μ to scale ν is a partial structure-preserving map*

$$\mathcal{R}_{\mu \rightarrow \nu} : D_\mu \dashrightarrow D_\nu$$

that relates descriptions across scale by coarse-graining, running, matching, rescaling, or scheme-compatible transformation. It is admissible as physical scale transport only on an overlap domain where witness comparison is defined.

Definition 6.3 (Overlap witness structure). *Two scale-indexed descriptions possess overlap witness structure when, after admissible coarse-graining, rescaling, redescription, or matching, their observables and empirical identifiers agree on a common reconstruction domain within the declared envelope, approximation, and truncation tolerances. The overlap is therefore not identity of microscopic variables, fields, or coordinate values. It is agreement of the witness-bearing content through which the descriptions can be re-identified as representing the same physical standing target.*

The partiality matters. Not every formal flow parameter is a physical continuation. RG transport is not an unrestricted map on all symbolic parameter assignments. It is defined only where the descriptions retain a witness relation, an admissibility envelope, and a quotient through which physical content can be compared.

Definition 6.4 (Standing-preserving scale transport). *An RG transport map $\mathcal{R}_{\mu \rightarrow \nu}$ is standing-preserving when it satisfies three conditions. Across such transport, what remains invariant is not any particular coordinate value, regulator, or scheme presentation, but the standing-bearing continuation structure through which overlap witnesses, quotient identity, and envelope-valid reconstruction continue to identify the same physical comparison target.*

- (S1) **Witness preservation on overlap:** *the descriptions possess overlap witness structure in the sense of Definition 6.3.*
- (S2) **Quotient factorization:** *if two presentations are equivalent in $\mathcal{Q}_\mu$, their images are equivalent in $\mathcal{Q}_\nu$ wherever the map is defined.*
- (S3) **Envelope respect:** *the map does not silently promote quantities outside their regime of validity, and any transition between envelopes is supplied with matching or boundary data.*

Definition 6.5 (Standing-bearing RG content). *Standing-bearing RG content is the quotient-level witness structure preserved under admissible scale transport. Equivalently, it is any scale-indexed feature whose loss would prevent empirical re-identification across the overlap domain: observable agreement, invariant scaling data, matching-preserved amplitudes, conserved charges, or other structures by which descriptions at different scales remain descriptions of the same physical content.*

These conditions state the minimal structure needed for scale-indexed descriptions to count as descriptions of the same physical standing content. They do not demand identity of fields, Lagrangian terms, parameter coordinates, regulator variables, or numerical coupling values.

Proposition 6.6 (Scale-indexed identity criterion). *Two scale-indexed descriptions can be treated as descriptions of the same standing-bearing physical content only if they are connected by standing-preserving scale transport.*

Proof. Physical identity across scale requires more than syntactic similarity. If witness preservation is absent, there is no empirical overlap on which identity can be checked. If quotient factorization is absent, equivalent presentations at one scale may become inequivalent at another, so the transported object is not well-defined. If envelope respect is absent, a parameter or expression may be used outside the domain in which it has standing. These cases exhaust the ways a scale comparison can lose admissible physical identity: loss of witness, loss of quotient, or loss of envelope. Hence same-standing scale comparison requires standing-preserving scale transport. $\square$

7 RG Continuation Loci and Target-Domain Retyping

RG notation often presents a path of values, for example $g(\mu)$ as μ varies, a family of Wilson coefficients $C_i(\mu)$, or a sequence of EFTs across thresholds. Such notation is indispensable, but it should not be confused with the structure that makes the path a physical continuation. A trace of values is not enough. The same standing-bearing RG target must be carried through the passage.

Definition 7.1 (RG continuation locus). *An RG continuation locus is the invariant role structure by which a scale-indexed family is treated as the same standing-bearing RG comparison target through admissible scale passage. It consists, as required by the case, of overlap witnesses, quotient identity, transport maps, envelope constraints, matching or boundary traces, and fail-closed conditions. It is not a hidden object beneath the flow; it is the role-structural locus through which an RG trajectory retains identity as a physical comparison.*

Definition 7.2 (Formal RG family). *A formal RG family is any indexed collection of couplings, schemes, cutoffs, EFT presentations, regulator choices, or parameter assignments. It may be written as $g(\mu)$, $\{D_\mu\}_{\mu \in M}$, $\{EFT_a\}_{a \in A}$, or a similar family. It is not automatically standing-bearing.*

Definition 7.3 (Explicitly gated RG target domain). *An explicitly gated RG target domain is a newly typed comparison domain whose membership rule, admissibility envelope, witness ledger, quotient relation, transport maps, matching data, and failure conditions are fixed before standing-bearing conclusions are drawn from a formal family.*

Lemma 7.4 (Trace of RG values is insufficient). *A formal RG family does not by itself define standing-preserving scale transport. It does so only when it preserves an RG continuation locus or is retyped as an explicitly gated RG target domain.*

Proof. A formal family records indexing, not standing-bearing identity. Without a continuation locus, it cannot distinguish same-flow transport from replacement, scheme relabeling, envelope drift, threshold failure, or unrelated coordinate succession. If the missing comparison structure is supplied by a larger family, that family now functions as the target domain and must be gated. Otherwise the family remains formal notation or exploratory modeling rather than physical RG continuation. $\square$

Proposition 7.5 (No inherited RG transport). *Standing-bearing RG transport is not inherited by formal multiplicity of scales, schemes, regulators, EFTs, or parameter assignments. It is licensed only by a fixed domain, an RG continuation locus, or an explicitly gated target-domain construction.*

Proof. Multiplicity supplies alternatives. It does not supply the admissibility relation by which those alternatives are compared as one standing-bearing physical target. If a fixed RG domain supplies overlap witnesses, quotient identity, transport, and envelope discipline, transport is evaluated there. If a larger comparison family supplies them, that family is retyped as a target domain. Without either structure, multiplicity is not a transport license. $\square$

Corollary 7.6 (Controlled scans remain admissible). *Parameter scans, scheme comparisons, EFT-family studies, phenomenological fits, and model-family analyses remain admissible modeling tools when their target domain and comparison predicates are explicit. The theorem constrains inherited physical standing, not conditional or exploratory usefulness.*

Proposition 7.7 (Target-domain retyping rule). *A cross-domain or family-level RG comparison that is not licensed inside the original fixed domain must be evaluated as a new target domain. If it is not retyped and gated in this way, it has no inherited standing-bearing RG authority.*

Proof. A cross-domain comparison changes the object being compared. If it preserves standing, it must fix the new admissibility envelope, witness ledger, quotient relation, transport maps, matching data, and failure conditions. That is exactly target-domain typing. If those structures are absent, there is no standing-bearing comparison; if they are present, the comparison is not inherited from the original domain but evaluated in the new one. $\square$

7.1 Continuous families and replicated sectors

Continuous RG families are legitimate when their flow is transport-closed: closed redescription loops cannot induce standing-distinguishable holonomy unless the family quotients into transport-invariant structure. Otherwise the putative continuous mobility is not a single admissible RG comparison class. Similarly, replicated or sector-multiplicity structures are not admitted merely by being duplicated. Symmetric redundancy is eliminable, unbounded standing-indistinguishable

duplication fails transport and non-degeneracy discipline, and any surviving multiplicity must be finite, non-degenerate, and role-certified inside the envelope.

This discipline does not alter Wilsonian practice. It states that running, universality, scheme comparison, and EFT families require a locus of continuation and a target domain of comparison. It is therefore a hardening of the RG thesis, not a replacement of RG machinery.

8 Raw Coupling Labels versus Standing-Bearing RG Content

A central test case is the running coupling. A coupling written $g(\mu)$ is often spoken of as if a single physical object changes its value with energy. This shorthand is useful, but it is not raw scalar identity. The value $g(\mu)$ is attached to a scale, scheme, normalization, and definition of the operator or vertex whose coefficient it is. Its standing depends on the RG structure connecting it to observables and to other scales.

Definition 8.1 (Raw coupling label). *A raw coupling label is a numerical coordinate $g_i(\mu)$ assigned to a term, operator, vertex, or effective parameter in a specified presentation at a specified scale. It is not by itself a standing-bearing invariant unless its role under witness comparison, quotient identity, envelope validity, and RG transport has been fixed.*

Definition 8.2 (RG transport class). *An RG transport class is an equivalence class of scale-indexed descriptions connected by standing-preserving RG maps. Its elements may have different raw coupling labels, field normalizations, operator bases, and scheme coordinates while preserving the same quotient-level physical standing on the relevant overlap.*

Lemma 8.3 (Raw coupling non-invariance). *A raw chart-numeric coupling value is generally not the invariant target of RG standing.*

Proof. A raw coupling value can change under scale evolution, under allowed reparameterization, under field normalization, or under scheme change. If the relevant witness ledger and quotient transport are preserved, such changes do not by themselves mark physical inequivalence. Conversely, identical raw numbers in two presentations need not identify the same physical content if the definitions, scales, schemes, or envelopes differ. Therefore raw numerical equality is neither necessary nor sufficient for RG standing. The invariant target is the transport class preserving witness and quotient structure. $\square$

This lemma is not a denial that couplings are physically meaningful. It is a role statement. Couplings are meaningful when they are embedded in the transport structure that specifies how they are measured, run, matched, compared, and used. The physical content is not a bare number but the role profile of that number inside RG transport.

Proposition 8.4 (Coupling role profile). *In an RG-governed domain, a coupling $g_i(\mu)$ typically has a composite role profile:*

$$g_i(\mu) : \quad \text{effective} + \text{transport-defined} + \text{presentation-dependent component},$$

with possible quotient-dependent or anchor-level content only after the domain and observable equivalence structure have been fixed.

Proof. The coupling is effective because its value is scale- and envelope-relative. It is transport-defined because its identity across scale is supplied by beta functions, matching maps, and RG

trajectories. It has a presentation-dependent component because scheme, basis, and normalization choices may alter the coordinate used to represent it. Only after quotient and witness structure are specified can one isolate which combinations, fixed-point data, or observable residues carry invariant load. Thus the role profile is composite rather than primitively scalar. $\square$

9 Overlap-Level Coincidence and Non-Divergence

The gauge-coupling analysis distinguishes two different targets: numeric coupling uniqueness and overlap-level coincidence. Numeric uniqueness asks whether admissibility forces a raw chart-numeric coupling to a unique value. It does not. Overlap-level coincidence asks whether two co-scoped descriptions can diverge on standing-bearing content while claiming to describe the same object-class on the same overlap. Such divergence is not a standing-preserving same-object comparison.

Definition 9.1 (Overlap standing domain). *An overlap standing domain is a region of comparison in which two scale-indexed descriptions claim authority over the same witness ledger, quotient structure, and object-class.*

Proposition 9.2 (Overlap non-divergence). *If two RG descriptions claim authority on an overlap standing domain, then their standing-bearing content must agree on that overlap. Any allowed difference must be absorbed by presentation change, envelope restriction, matching, or transport.*

Proof. Co-authority on an overlap means that both descriptions claim to represent the same physical content for the same witness ledger and quotient object. If their standing-bearing content differed on that overlap, the same witness target would receive incompatible physical identities. The comparison would then cease to be a same-object claim. Therefore admissible differences must be classified as presentation-dependent, effective-envelope-relative, or transport-mediated. Otherwise the descriptions are not co-scoped on the stated overlap. $\square$

This proposition is weaker than numerical uniqueness and stronger than unconstrained pluralism. It does not require that all coordinate values agree. It requires agreement of standing-bearing content where the descriptions claim the same physical authority. This is the role played by matching conditions, scheme transformations, and universal quantities in ordinary RG reasoning.

10 The RG Transport Proposition

The following local proposition separates admissible running from unlicensed parameter reassignment.

Proposition 10.1 (RG transport proposition). *Let D_μ and D_ν be scale-indexed descriptions connected by an RG map $\mathcal{R}_{\mu \rightarrow \nu}$. If the map preserves the witness ledger on overlap, factors through the standing quotients, and respects the effective envelopes, then differences among raw parameters at μ and ν are admissible scale-transport differences rather than unlicensed same-domain reassignment.*

Proof. By witness preservation, the descriptions agree on the empirical content used for comparison. By quotient factorization, equivalent presentations at the source scale remain equivalent at the target scale, so physical identity does not depend on representative choice. By envelope respect, quantities are not promoted outside their valid domain without matching. Therefore parameter differences induced by the map express lawful scale transport within the admissible comparison class. They are not free replacement of constants inside a fixed domain, because the map supplies the transport relation by which the altered coordinates retain standing. $\square$

Corollary 10.2 (RG running is not a counterexample to fixity). *RG running does not refute the fixed-domain claim that standing-bearing constants are not freely variable same-domain parameters. Running is admissible variation under a transport law; unlicensed same-domain reassignment lacks that transport basis.*

Proof. The fixed-domain result targets variation without standing-preserving transport. RG running, when admissible, is precisely a standing-preserving transport process. The two cases are therefore distinct. $\square$

Remark 10.3. *The point is not that RG flow makes constants “unchanged.” The point is that the relevant sameness is quotient-level standing, not coordinate-level numerical equality. RG flow can preserve physical identity while changing the coordinates used to present it.*

11 Scheme Dependence as Presentation-Dependent Skin

Scheme dependence is sometimes treated as an embarrassment: if beta functions or coupling values depend on choices, which quantity is real? The admissibility answer is straightforward. A scheme coordinate is presentation-dependent when changing it preserves the witness ledger and quotient-level physical content. It becomes physically load-bearing only when a difference in scheme is tied to a difference in observables, envelopes, or matching conditions.

Definition 11.1 (Presentation-dependent RG structure). *An RG structure is presentation-dependent when admissible changes of scheme, regulator, field normalization, operator basis, or coordinate chart alter its formal expression while preserving all standing-bearing witness and quotient content on the relevant domain.*

Lemma 11.2 (Scheme-skin lemma). *A renormalization scheme is presentation-dependent skin exactly to the extent that changes of scheme preserve the standing-bearing witness ledger and the quotient transport class.*

Proof. If a scheme change alters only coordinates, labels, subtraction convention, or normalization while leaving observable comparison and transport equivalence intact, the change has no independent standing-bearing authority and is presentation-dependent. If the change alters the witness ledger, matching relation, or domain of validity, it is no longer presentation-dependent scheme skin; it changes the effective envelope or physical content. The criterion therefore lies in witness and quotient preservation, not in whether the formal expression changes. $\square$

11.1 Universal and non-universal beta-function data

The familiar distinction between universal and scheme-dependent RG data is naturally expressed in this vocabulary. Some coefficients, critical exponents, anomalous dimensions at fixed points, or scaling relations may remain invariant under admissible reparameterization and function as quotient-level tensor content. Other coefficients or coordinate expressions depend on scheme and function as presentation-dependent. The same calculation may contain both.

Proposition 11.3 (Universal data criterion). *An RG quantity is universal, in the admissibility sense, when it is invariant under the admissible presentation changes of the comparison class and remains recoverable from the standing-bearing witness ledger. Otherwise it is presentation-dependent or envelope-relative.*

Proof. Universality requires invariance under the admissible changes that do not alter physical standing. If a quantity changes under such changes while witnesses are preserved, it is not tensor-level universal data. If it remains invariant under those changes and is tied to observable or reconstructible content, it carries quotient-level standing. This is exactly the tensor/skin distinction applied to RG descriptions. $\square$

11.2 Regulators and cutoffs

A regulator can have several roles. It may be presentation-dependent when it renders intermediate expressions finite without changing witness content. It may mark an effective envelope when it records the scale beyond which a description is not admissible. It may represent physical threshold structure if tied to real degrees of freedom or matching data. These roles must be kept distinct.

Table 3: Role discipline for common RG presentation structures.

Object	Role when witnesses are preserved	Role change when witnesses are not preserved
Subtraction scheme	Presentation-dependent coordinate	Different finite parts may require matching; otherwise no physical difference.
Field normalization	Presentation-dependent chart choice	If observable matrix elements or source couplings change without compensation, quotient mismatch.
Regulator	Presentation-dependent device or envelope marker	Physical threshold only if tied to witness-bearing degrees of freedom.
Cutoff	Envelope boundary or calculational skin	Physical interpretation only with boundary or witness justification.
Operator basis	Presentation-dependent basis of effective terms	Transport-defined if basis reconstruction carries physical content.
Renormalization scale μ	Scale coordinate for transport	Not a physical selector unless tied to envelope, matching, or observable scale.

12 Effective Field Theory as Envelope-Bounded Admissibility

Effective field theory is the most direct physics-facing example of admissibility discipline. An EFT is a controlled description admissible within an operational envelope, equipped with matching data, power counting, and a witness ledger. Its coefficients can be real physical parameters in that envelope without becoming primitive all-scale constants. Envelope-bounded validity does not imply physical inadequacy; it specifies the admissibility conditions under which the effective description preserves standing.

Definition 12.1 (EFT envelope). *An EFT envelope is a tuple*

$$\mathcal{E}_{\text{EFT}} = (\Lambda, \mathcal{O}, \mathcal{P}, \mathcal{M}, \epsilon),$$

where Λ marks the cutoff or breakdown scale, $\mathcal{O}$ the operator basis, $\mathcal{P}$ the power-counting or relevance ordering, $\mathcal{M}$ the matching data to a higher or adjacent description, and ϵ the tolerated approximation or error regime.

Lemma 12.2 (EFT envelope lemma). *An EFT coefficient has admissible physical standing only relative to its envelope, matching conditions, and witness ledger.*

Proof. A Wilson coefficient or low-energy constant receives its operational meaning from the operator basis in which it appears, the cutoff or scale at which it is defined, the matching or fitting conditions that fix it, and the observables it predicts within the envelope. Removing those structures leaves a raw numerical symbol without a complete physical role. Therefore the coefficient is physical as an envelope-bound object, not as an unrestricted primitive scalar. $\square$

Proposition 12.3 (EFT admissibility proposition). *An EFT is an admissible physical description when its envelope, matching conditions, and witness ledger are explicit enough to preserve standing over its intended domain. Promotion of EFT parameters outside that domain is scope transport requiring new admissibility data.*

Proof. Within the envelope, observables are defined, coefficients are calibrated, and approximation error is controlled. This supplies witness preservation and operational standing. Outside the envelope, omitted operators, thresholds, or degrees of freedom may become standing-bearing. The EFT parameters therefore do not retain their interpretation outside the envelope by symbolic continuation alone. A lawful extension requires matching or a new domain. $\square$

This proposition is not a criticism of EFT. It identifies the structural reason EFT is powerful. EFT separates what must be retained from what is suppressed, organizes uncertainty by relevance, and preserves standing within a controlled regime. In the present terminology, EFT is a disciplined envelope of admissible description.

The added admissibility claim is specific. It supplies criteria for when EFT continuation ceases to preserve standing: when a cutoff is promoted beyond its envelope without matching, when a regulator is treated as physical without witness preservation, when a UV extrapolation is made without a transport map, or when a threshold is crossed without boundary data. These do not indict EFT methodology. They identify incorrect role assignment for an EFT object.

Corollary 12.4 (EFT continuation criterion). *An EFT continuation remains standing-preserving only while its witness ledger, quotient relations, and envelope controls are maintained or replaced by explicit matching data. Beyond that point the continuation is a new admissibility problem rather than a mere extension of the same parameter role.*

12.1 Wilson coefficients and low-energy constants

Wilson coefficients are often the most visible example of effective role structure. They are meaningful, predictive, and empirically constrained. But their meaning is not independent of scale, basis, matching, and truncation. A coefficient $C_i(\mu)$ is not an all-scale physical primitive. It is an effective coordinate whose physical load is transported through RG equations and matching relations.

Proposition 12.5 (Wilson coefficient role profile). *A Wilson coefficient in an EFT is generally effective and transport-defined, with presentation-dependent components determined by basis and scheme. It carries quotient-level invariant content only through observable combinations or universal transport residues.*

Proof. The coefficient is effective because it is meaningful within a cutoff and truncation. It is transport-defined because its value at different scales is related by RG flow. Basis and scheme choices alter its coordinate expression, adding presentation dependence. Observable amplitudes, matching-invariant combinations, or universal scaling data are the places where quotient-level content may be isolated. Hence the profile is composite. $\square$

13 Universality Classes as Quotient Persistence

Universality is one of the clearest illustrations that RG flow is not simple coordinate accounting. Different microscopic systems can share critical exponents, scaling functions, and fixed-point structure. From the present perspective, a universality class is the quotient object that remains after non-standing-bearing microscopic detail is removed by admissible coarse-graining.

Definition 13.1 (Universality quotient). *A universality quotient is an equivalence class of microscopic or effective descriptions whose admissible coarse-graining maps factor through a common standing quotient on the relevant witness overlap. The quotient identifies descriptions that differ in presentation or irrelevant microscopic detail while retaining the same universal witness structure.*

Theorem 13.2 (Universality quotient theorem). *Two microscopically distinct descriptions belong to the same universality class if and only if their admissible coarse-graining maps factor through the same standing quotient on overlap witness structure.*

Proof. Suppose two descriptions belong to the same universality class. By ordinary RG practice, their non-universal microscopic details do not affect the critical witness profile: critical exponents, scaling relations, and other universal data agree on the relevant overlap witness structure. The coarse-graining maps therefore identify the same standing-bearing profile after quotienting away presentation and irrelevant microscopic surplus. They factor through a common standing quotient.

Conversely, suppose the admissible coarse-graining maps of two descriptions factor through the same standing quotient on the relevant overlap witness structure. Then the structures that remain after coarse-graining are the same structures by which critical behavior is identified: scaling dimensions, exponents, invariant amplitude relations, or other universal witness data appropriate to the domain. Differences outside that quotient are not standing-bearing for the universal regime. The descriptions are therefore in the same universality class. Hence universality is not only resemblance among microscopic systems; it is quotient persistence under admissible RG transport. $\square$

Corollary 13.3 (Microscopic surplus). *Microscopic detail not retained in the universality quotient is surplus relative to the universal standing target, though it may remain physical in another envelope or for non-universal observables.*

Proof. A detail may be irrelevant to critical standing while remaining meaningful elsewhere. Its loss under the universal quotient does not make it unreal; it means it does not carry the standing-bearing content of that universality class. The role is envelope-relative. $\square$

13.1 Universality and quotient convergence

The usual success of universality is sometimes described as the surprising fact that many different microscopic systems behave alike at criticality. The admissibility account identifies the structural reason for this convergence. Coarse-graining discards or suppresses presentation and irrelevant envelope detail while preserving a quotient-level standing profile. Once the quotient is fixed, multiple representatives can share the same universal content without sharing microscopic identity.

Remark 13.4. *This interpretation does not replace the calculation of universality classes. It says what kind of object the calculation identifies: a quotient-level witness profile preserved under RG transport.*

14 Fixed Points as Transport-Stable Quotient Loci

A fixed point is usually introduced as a point g_* in coupling space for which

$$\beta(g_*) = 0.$$

This is correct as a coordinate condition, but the structural significance of a fixed point is broader. A fixed point has physical authority only when RG evolution preserves the relevant standing quotient, admits invariant overlap observables, and retains admissible reconstruction under allowed presentation changes.

Definition 14.1 (Transport-stable fixed point). *A fixed point is transport-stable when RG transport preserves the standing quotient under scale evolution on the relevant envelope. Operationally, this requires: coordinate stationarity in an admissible chart, preservation of the witness profile under scale transformations, and stability of the quotient data under allowed scheme or normalization changes.*

Proposition 14.2 (Fixed-point quotient-stability proposition). *A admissible RG fixed point is a transport-stable quotient locus: a locus at which the standing quotient, not only a coordinate value, is preserved by the relevant RG transformation.*

Proof. The equation $\beta(g_*) = 0$ states coordinate stationarity in a chosen parametrization. Physical standing requires additional structure. The stationary coordinate must correspond to a scale-invariant witness profile, and equivalent presentations must induce the same quotient data. When these conditions hold, the fixed point preserves the standing quotient under RG evolution. When they do not hold, the coordinate zero is a formal feature of a chart without independent standing-bearing authority. Thus the admissible fixed point is the quotient locus that retains admissible reconstruction under RG evolution, rather than only the zero of a displayed beta function. $\square$

14.1 UV and IR roles

Fixed points can play different roles.

- An IR fixed point may be an attractor of low-energy effective standing, explaining why many UV details are not visible in the IR quotient.
- A UV fixed point may serve as a possible high-scale admissibility anchor, but only if it supplies standing-preserving continuation rather than a formal endpoint alone.
- A Gaussian fixed point preserves a free-theory quotient profile; an interacting fixed point preserves nontrivial quotient structure.
- A fixed line or surface may carry transport-stable structure with marginal directions requiring further role analysis.

The classification is role-relative. The same mathematical fixed point can be a calculational tool, a universal standing profile, an envelope anchor, or a formal artifact depending on whether overlap observables, quotient structure, and admissible reconstruction are preserved.

Corollary 14.3 (Fixed point endpoint discipline). *A formal fixed point is not, by itself, an admissible physical endpoint. It must carry or connect to standing-bearing witness and quotient structure.*

Proof. Coordinate stationarity alone supplies neither empirical identification nor envelope continuation. Without witness and quotient structure, the fixed point has formal role but not physical endpoint authority. With such structure, it may function as a quotient-stable locus or anchor. The role is not automatic. $\square$

15 Relevant, Irrelevant, and Marginal Operators as Role Diagnostics

Operator relevance is often presented by scaling dimension. The admissibility interpretation treats relevance as a transport-role diagnostic: how an operator’s coefficient behaves under RG transport relative to the standing target.

Definition 15.1 (Transport-role diagnostic). *An operator classification is a transport-role diagnostic when it determines whether a perturbation changes the admissible low-energy quotient, decays into presentation or microscopic surplus, remains scale-sensitive, or requires additional matching data.*

Proposition 15.2 (Operator role proposition). *Relevant, irrelevant, and marginal operators classify how perturbations affect standing under scale transport.*

Proof. A relevant perturbation grows under flow and can alter the effective envelope or low-energy quotient, so it often requires boundary input. An irrelevant perturbation decays relative to the target quotient and may become microscopic surplus for the low-energy standing profile. A marginal perturbation requires higher-order or nonperturbative transport analysis to determine whether it changes standing. The classification therefore describes the role of an operator in preserving or altering the quotient under scale. $\square$

Remark 15.3 (Dangerously irrelevant operators). *A dangerously irrelevant operator shows why the role diagnostic is not reducible to dimensional classification alone. An operator may be irrelevant near one fixed point and still affect ordered phases, amplitudes, or other witness-bearing quantities. In the present vocabulary, it is not pure presentation-dependent skin solely because it is irrelevant by dimension. Its role must be evaluated through the witness ledger and envelope.*

Table 4: Operator classes as admissibility roles.

Operator class	RG behavior	Admissibility interpretation
Relevant	Grows toward target scale	May alter envelope trajectory or low-energy quotient; usually demands boundary input.
Irrelevant	Decays relative to target	Often microscopic surplus for the target quotient, but not globally unreal.
Marginal	Requires further transport analysis	May be exactly stable, marginally relevant, or marginally irrelevant depending on beta structure.

Operator class	RG behavior	Admissibility interpretation
Dangerously irrelevant	Decays locally but affects other witnesses	Cannot be treated as skin without checking witness and phase structure.
Redundant	Generated by field redefinition or equations of motion	Presentation-dependent if witnesses are preserved.

16 Thresholds, Matching, and Boundary-Trace Fixation

RG flow across thresholds is not continuation of the same formula alone. When degrees of freedom become inactive, new descriptions are introduced, or envelopes change, standing must be transferred. Matching supplies the bridge.

Definition 16.1 (Boundary trace for RG matching). *A boundary trace for RG matching is a declared, witness-preserving relation at an interface between effective domains that fixes how standing-bearing quantities in one envelope are inherited by another.*

Proposition 16.2 (Threshold matching proposition). *Threshold matching is boundary-trace fixation for RG regime transition. A new effective envelope inherits admissible standing only through lawful witness-preserving matching.*

Proof. Crossing a threshold changes the degrees of freedom or effective description. Without matching, quantities in the lower-energy description need not preserve the observables or quotient relations of the higher-energy description. Matching fixes boundary data at the interface, declaring how amplitudes, correlators, charges, coefficients, or observables agree within tolerance. This is exactly the function of a boundary trace: it transfers standing without importing a hidden selector. $\square$

Proposition 16.3 (Matching continuity criterion). *Threshold matching preserves standing only if overlap observables admit admissible reconstruction across the boundary trace. Equivalently, amplitudes, correlators, charges, or other witness-bearing quantities computed on the two sides must agree within the declared envelope and truncation error.*

Proof. A threshold changes the descriptive basis: degrees of freedom may be integrated out, operators may be reorganized, and coefficients may be redefined. If no overlap observable can be reconstructed across the interface, then the lower-energy description has not inherited the standing of the higher-energy one. Matching equations provide the reconstruction relation. They are therefore continuity conditions for standing across envelopes, not discretionary labels. $\square$

A schematic heavy-threshold example illustrates the point. Let a full theory contain a heavy degree of freedom of mass M , and let the low-energy EFT be used for external momenta $p \ll M$. Matching requires relations of the form

$$\mathcal{A}_{\text{full}}(p; M, \mu) = \mathcal{A}_{\text{EFT}}(p; C_i(\mu), \mu) + O\left((p/M)^{n+1}\right),$$

for the chosen truncation order. The equality is not identity of Lagrangians or fields. It is reconstruction of the witness ledger within the declared envelope. The coefficients $C_i(\mu)$ acquire standing because they are fixed by this boundary trace.

16.1 Matching as a transport condition

Treating matching as pure notation obscures its role. When matching carries witness-preserving boundary data, it is a transport condition rather than a presentational convenience. The coefficient values of the lower-energy EFT are not inherited by notation; they are fixed through a boundary relation that preserves observable content across envelopes.

Corollary 16.4 (No silent decoupling). *Dropping degrees of freedom does not by itself preserve standing. Decoupling requires a transport or matching relation showing which witness-bearing effects are preserved in the new envelope.*

17 Dimensional Transmutation and Scale Anchors

Dimensional transmutation is a central case where role type changes. A dimensionless running coupling can be replaced, under the RG structure, by a dimensional scale such as Λ_{QCD} . This is not numerical creation from nothing. It is a change in admissibility role: a coupling coordinate is converted into a transport-defined scale anchor relative to a boundary condition.

Proposition 17.1 (Transmutation role proposition). *Dimensional transmutation transforms a coupling-coordinate description into a scale-anchor description within an RG transport class. The resulting scale is effective, transport-defined, and boundary-sensitive rather than a raw primitive scalar.*

Proof. In an asymptotically free theory, specifying the value of a running coupling at one scale can be equivalently expressed by specifying a scale at which the coupling reaches a conventionally defined strength. The scale depends on the RG equation, scheme convention, and boundary input. It is therefore not a freely floating primitive. It functions as an anchor for the effective regime by encoding transport history in a scale parameter. $\square$

17.1 The QCD scale as example

The QCD scale illustrates the point. Λ_{QCD} is not a dimensionful constant inserted into the theory independently of RG structure. It is tied to the running of the strong coupling, the renormalization scheme, and the scale at which perturbative descriptions leave their intended envelope or confinement-relevant physics becomes dominant. Its role is composite: effective, transport-defined, boundary-sensitive, and quotient-relevant through hadronic observables.

Remark 17.2. *The role classification does not diminish the physical importance of Λ_{QCD} . It clarifies why its physical importance is not exhausted by treating it as a raw number. Its standing comes through transport, matching, and observable scale setting.*

18 AMetric Boundary and No Default RG Selector

The AMetric boundary result in the AASC corpus supplies a discipline for selector claims. Before admissibility is fixed, there is no canonically admissible metric, ordering, representative, endpoint, or measure over the relevant comparison space. Applied to RG, this means that no scale, scheme, regulator, cutoff, fixed point, UV endpoint, or trajectory is canonically privileged by formal nameability alone.

Proposition 18.1 (No default RG selector). *Prior to admissibility fixation, there is no canonical RG selector for scale, scheme, cutoff, regulator, representative, trajectory, fixed point, or endpoint. Any such selection requires support from witness structure, quotient preservation, envelope conditions, or explicitly declared boundary data.*

Proof. A selector that chooses one representative among many RG presentations must either be presentation-dependent, witness-bearing, quotient-preserving, envelope-fixing, or externally stipulated. If it is presentation-dependent, it has no independent physical authority. If it is witness-bearing or envelope-fixing, its authority comes from admissibility data. If it is externally stipulated without such data, it imports a hidden preference. Therefore no default selector is available before admissibility conditions are fixed. $\square$

This proposition does not prohibit common RG conventions. Choosing μ near a characteristic physical scale, selecting a convenient subtraction scheme, or using a regulator is often practically necessary and scientifically legitimate. The proposition only blocks promotion of such choices into canonical explanations when they lack standing-bearing support.

Corollary 18.2 (Parameter scans and explanatory authority). *RG parameter scans may be useful exploratory tools, but their explanatory authority depends on the admissibility of the measure, range, scheme, and witness relation used in the scan.*

19 Null-Regime Exclusion for RG Endpoints

The null-regime constraint is used here in a restricted way. The claim is not that RG endpoints force existence or that every mathematical endpoint is realized. The claim is narrower: an endpoint with no witness, quotient, transport, or envelope structure lacks the resources to function as an admissible physical explanation.

Definition 19.1 (Null RG endpoint). *A null RG endpoint is a purported endpoint, completion, or limit that lacks witness ledger, quotient structure, admissible transport, and effective envelope data while being used as if it explained physical standing.*

Proposition 19.2 (Null endpoint exclusion). *A null RG endpoint is an admissible physical endpoint only if standing-bearing structure is preserved or transferred to it by lawful transport.*

Proof. A physical endpoint must be capable of preserving or terminating a standing-bearing comparison in a well-defined way. If no witness ledger is present, there is no empirical content. If no quotient is present, there is no identity relation. If no transport is present, there is no lawful continuation. If no envelope is present, there is no domain of validity. Such an endpoint may be a formal limit, but it does not by itself supply a physical explanation of the standing-bearing structure from which it was reached. $\square$

Remark 19.3. *This result applies symmetrically to vacuous UV completions, formal IR endpoints, and regulator limits treated as physical without witness transport. It does not deny the usefulness of formal limits; it constrains their interpretive role.*

20 The RG Standing Preservation Theorem

We now assemble the preceding pieces into the main theorem.

Theorem 20.1 (RG standing preservation theorem). *Let D_μ and D_ν be scale-indexed admissibility domains connected by an RG map $\mathcal{R}_{\mu \rightarrow \nu}$. Suppose that:*

- (H1) $\mathcal{R}_{\mu \rightarrow \nu}$ preserves overlap witness structure;
- (H2) $\mathcal{R}_{\mu \rightarrow \nu}$ factors through the standing quotients $\mathcal{Q}_\mu$ and $\mathcal{Q}_\nu$;
- (H3) $\mathcal{R}_{\mu \rightarrow \nu}$ respects the effective envelopes or supplies lawful matching data between them;
- (H4) changes of scheme, regulator, field normalization, and coordinate representation are either quotient-preserving presentation changes or are explicitly assigned envelope/witness load;
- (H5) any endpoint or fixed locus used in the comparison carries standing-bearing witness, quotient, transport, or envelope structure;
- (H6) the scale-indexed relation preserves an RG continuation locus, or else the comparison is explicitly retyped as a gated RG target domain.

Then scale-dependent parameter change induced by $\mathcal{R}_{\mu \rightarrow \nu}$ is admissible transport rather than unlicensed same-domain variation. The standing-bearing content of the RG relation is the induced quotient transport class and its continuation locus, not the raw coordinate values of the running parameters.

Proof. Assume the hypotheses. By (H1), the source and target descriptions possess overlap witness structure on which physical identification and reconstruction are possible. By (H2), equivalent representatives at the source are mapped to equivalent representatives at the target, so the transport is well-defined on standing content rather than chart choice. By (H3), the flow does not silently apply effective quantities outside their domain; envelope changes are mediated by matching. By (H4), presentational changes are separated from standing-bearing changes, preventing scheme or regulator coordinates from being mistaken for invariant load. By (H5), endpoints and fixed loci used in the comparison are not vacuous selectors. By (H6), the flow is not a mere trace of values: it either preserves the same standing-bearing RG continuation locus or is evaluated in a newly typed target domain.

Now consider a raw parameter difference between the descriptions. If the difference is generated by the RG map while the above conditions hold, then it is part of the lawful scale transport relation. It is not unlicensed same-domain reassignment, because its legitimacy depends on the transport structure. If a raw difference changes witnesses, lacks quotient factorization, violates envelope conditions, relies on a null endpoint, or fails to preserve an RG continuation locus, then it falls outside admissible RG continuation under the hypotheses unless it is retyped as a new target-domain comparison. Hence admissible RG flow preserves standing by quotient transport, and the invariant target is the transport class rather than raw numerical sameness. $\square$

Corollary 20.2 (RG as exemplar of admissibility discipline). *Renormalization is not an exception to standing fixity. It is one of the clearest physical examples of standing-preserving admissibility discipline.*

Proof. Standing fixity denies free same-domain variation without transport. RG supplies exactly the transport structure needed for scale-dependent variation to retain physical standing. Therefore RG exemplifies the discipline rather than contradicting it. $\square$

20.1 The theorem ladder

For reference, the argument can be organized as the following theorem ladder.

Table 5: The RG theorem ladder.

Result	Function in the argument
Scale-indexed domain definition	Specifies the comparison object for RG standing.
Overlap witness structure definition	Specifies the common empirical reconstruction domain for RG comparison.
Standing-bearing RG content definition	Identifies the quotient-level witness structure preserved under admissible scale transport.
RG continuation locus definition	Identifies what persists through formal flow notation as the same standing-bearing comparison target.
Trace of RG values insufficiency lemma	Separates formal scale-indexing from physical RG continuation.
No inherited RG transport proposition	Blocks transport claims based on formal multiplicity alone.
Target-domain retyping rule	Explains how cross-domain or family-level RG comparison becomes a new gated comparison domain.
Raw coupling non-invariance lemma	Separates coupling coordinates from invariant physical content.
Scheme-skin lemma	Classifies scheme and regulator dependence as presentation-dependent when witnesses are preserved.
EFT envelope lemma	Binds effective coefficients to operational domains and matching.
Universality quotient theorem	Identifies universality classes by common factorization through the same standing quotient.
Fixed-point quotient-stability proposition	Interprets fixed points as loci preserving standing quotients under RG evolution, not only coordinate zeros.
Threshold matching proposition	Treats regime transition as boundary-trace fixation.
Matching continuity criterion	Requires overlap-observable reconstruction across threshold boundaries.
No default RG selector	Blocks unsupported scale, scheme, endpoint, or representative selection.
Null endpoint exclusion	Prevents vacuous RG endpoints from functioning as physical explanations.
RG standing preservation theorem	Concludes that admissible RG flow is standing-preserving quotient transport.

21 Error Taxonomy

The admissibility account is useful because it diagnoses recurring interpretive errors in discussions of RG.

Table 6: Common RG role errors.

Error	Correction
Treating running couplings as freely variable scalar constants	Running is transport-defined; the coupling coordinate has meaning only inside a scale, scheme, and witness relation.
Treating scheme coordinates as tensor load	Scheme dependence is presentation-dependent when witnesses and quotient content are preserved.
Treating EFT coefficients as primitive constants	EFT coefficients are envelope-bound and matching-dependent.
Treating cutoff dependence as ontology	A cutoff is physical only when tied to witness-bearing threshold or boundary structure.
Treating universality as coincidence	Universality is quotient persistence under coarse-graining and scale transport.
Treating fixed points as physical endpoints by default	Fixed points require witness, quotient, and envelope role to carry physical endpoint authority.
Treating threshold matching as pure notation	Matching transfers standing across envelope boundaries.
Treating UV completion as same-domain variation	UV completion often changes anchors, envelopes, or witness structure; it is not, by itself, a variation inside the old domain.
Treating irrelevant operators as globally unreal	Irrelevance is target-relative; other envelopes may retain standing-bearing effects.
Treating regulator removal as physical proof by itself	A limit has physical authority only if standing-bearing content is preserved through the limiting process.

21.1 Why these errors recur

They recur because RG compresses several roles into the same notation. The same symbol may mark a coordinate, an effective parameter, a boundary condition, a transport history, or an observable combination. Without role discipline, one can move between these meanings without noticing the scope change. The resulting tension is familiar: a coupling both “changes” and “is a constant,” a cutoff both “does not matter” and “sets the theory,” a fixed point both “is a coordinate zero” and “explains universality.” The admissibility account preserves the physics behind these statements while assigning each statement to the role in which it is true.

22 Worked Example: Gauge Coupling under RG

Consider a single gauge coupling with beta function

$$\mu \frac{dg}{d\mu} = \beta(g).$$

This equation by itself is not yet a complete physical object. Its admissibility role depends on the definition of g , the renormalization scheme, the scale range, the observable used to fix the boundary condition, and the envelope in which the description is valid.

22.1 Role decomposition

Table 7: Role decomposition for a gauge coupling under RG.

Object	Admissibility role	Interpretation
$g(\mu)$	Effective / transport-defined coordinate	Scale-attached coupling label; meaningful through scheme, boundary condition, and observable relation.
$\beta(g)$	Transport rule	Specifies how the coordinate changes with scale in a chosen presentation.
Scheme choice	Presentation- dependent when witnesses preserved	Alters coordinate representation without changing physical standing.
Boundary condition $g(\mu_0)$	Boundary/envelope input	Fixes trajectory relative to a measurement or matching datum.
Fixed point $\beta(g_*) = 0$	Quotient-stable RG locus	Physically load-bearing only if tied to quotient and witness structure.
RG trajectory	Transport-defined standing carrier	Represents the admissible scale relation, not a set of unsupported constant assignments.
Λ from transmutation	Effective scale anchor	Encodes transport history and boundary input in a scale parameter.
Observable amplitude	Witness ledger component	Provides empirical standing by which the coupling coordinate is calibrated.

The table shows why RG running is not unconstrained variation. Replacing $g(\mu)$ by another number without changing the trajectory, boundary data, scheme definition, or witness relation is not a well-formed physical comparison. Conversely, a change in $g(\mu)$ generated by the beta function is admissible when it is part of the standing-preserving transport class.

22.2 Trajectory, scheme, fixed point, and transmutation

At one loop, an asymptotically free schematic beta function may be written

$$\mu \frac{dg}{d\mu} = -b_0 g^3 + O(g^5), \quad b_0 > 0.$$

The displayed equation is a transport rule in a chosen coordinate. It does not say that the isolated number g is the invariant content. A boundary condition at μ_0 selects a trajectory; a scheme choice fixes the coordinate presentation; and a witness observable calibrates the boundary condition. The same physical standing may be represented by a transmutation scale of the form

$$\Lambda \sim \mu \exp \left[-\frac{1}{2b_0 g^2(\mu)} \right]$$

Table 8: What changes and what is preserved in admissible gauge-coupling RG transport.

Changes under admissible RG description	Preserved standing-bearing content
Raw coupling labels such as $g(\mu)$	Quotient-level witness structure and calibrated observable content
Scheme coordinates and subtraction conventions	Overlap observables and admissible reconstruction relations
Regulator details or field normalizations	Universal data, conserved charges, and transport-equivalent amplitudes
Effective coefficients across scale or threshold	Matching-preserved boundary traces and RG transport class

up to scheme-dependent factors and higher-order corrections. The role has changed: a running coupling coordinate has been traded for a scale anchor encoding the trajectory and boundary input.

A fixed point enters differently. If $\beta(g_*) = 0$, then g_* is a stationary coordinate in the chosen scheme. It becomes standing-bearing only when the associated scaling profile is invariant under admissible redescription and visible in the witness ledger. Thus the fixed point is not the number alone; it is the transport-stable quotient profile carried by scaling dimensions, exponents, or other invariant data.

Finally, an observable $O(Q)$ computed at physical scale Q may be represented using a renormalization scale μ :

$$O(Q) = F(Q/\mu, g(\mu), C_i(\mu), \dots).$$

Physical standing requires that admissible changes of μ and scheme leave the witness value invariant within the envelope. In schematic form, this is the role of RG invariance: the coordinate dependence introduced by renormalization is compensated by transport of couplings and coefficients. The witness, not the coordinate, is the empirical ledger.

22.3 Three comparisons

There are three distinct comparisons.

Same trajectory, different scale. Values $g(\mu)$ and $g(\nu)$ differ, but the beta function and boundary data connect them. This is admissible transport.

Same physics, different scheme. Values $g_A(\mu)$ and $g_B(\mu)$ may differ because the coupling coordinate differs. If observables and quotient transport agree, the difference is presentation-dependent.

Different boundary input or theory. Values may differ because the empirical boundary condition, field content, gauge structure, or matching data differ. This is not running alone; it may be a different envelope, domain, or theory.

Proposition 22.1 (Gauge-coupling interpretation). *A gauge coupling under RG carries physical standing through its role in a transport class calibrated by witness data, not through raw scale-independent numerical identity.*

Proof. The coupling coordinate changes with scale, may change with scheme, and requires boundary input. Its physical role is recovered by the RG relation and its observable calibration. Therefore standing belongs to the transport class and witness relation, not to raw numerical sameness. $\square$

23 Objection Matrix

Table 9: Objections and replies.

Objection	Reply
RG already shows constants vary.	RG shows lawful scale transport, not unlicensed same-domain reassignment. The variation is generated by a transport rule preserving witness and quotient structure.
This adds no new beta functions.	Correct. The contribution is structural, not dynamical. It classifies what RG flow preserves and what kind of variation it licenses.
Scheme dependence is physically meaningful.	It can be, if tied to witness, envelope, or matching differences. When witnesses are preserved, scheme dependence is presentation-dependent.
EFT parameters are real.	Yes, within their envelope. The claim is not unreality but role discipline: EFT coefficients are envelope-bound and matching-dependent.
Fixed points are already understood.	The analysis does not replace the standard account. It identifies the admissibility role of fixed points as loci preserving standing quotients under RG evolution.
Universality is already explained by RG.	Yes. The analysis identifies the kind of object isolated by RG explanation: a quotient-persistent standing profile.
Regulators are not always presentation-dependent.	Correct. The role depends on whether the regulator is a calculational device, envelope marker, or witness-bearing physical threshold.
UV completion changes the theory.	Often true. The present framework treats that as anchor or envelope change, not same-domain variation.
The framework is philosophical rather than physical.	The argument is about the structural preconditions of physical comparison: witness preservation, quotient identity, scale transport, and envelope validity. These are ordinary RG requirements.
Parameter scans remain useful.	Yes. The theorem constrains canonical explanatory authority, not exploratory or phenomenological usefulness. Controlled scans inside explicitly governed target domains remain admissible modeling tools.
One can always define a larger RG family.	Yes, but if the larger family carries standing-bearing comparison, it is a newly gated target domain rather than inherited transport from multiplicity alone.

Objection	Reply
Continuation-locus language adds ontology.	No. The continuation locus is the role structure preserving RG identity through transport; it is not a hidden substance or physical object.

24 Conclusion

RG flow is not a counterexample to standing fixity. It is one of its clearest physical exemplars. Renormalization teaches that physical identity across scale is not raw numerical sameness. Couplings run, schemes vary, regulators appear and disappear, degrees of freedom are matched across thresholds, irrelevant operators decay, relevant operators grow, fixed points preserve quotient structure when their data are admissible under the relevant witness and envelope conditions, and universality classes retain common overlap observables. The standing-bearing content is the quotient-level transport profile preserved through these changes.

The admissibility interpretation therefore reframes RG objects as follows. Running couplings are effective, transport-defined coordinates. Scheme choices are presentation-dependent when witness structure is preserved. EFTs are envelope-bounded admissible descriptions. Universality classes are quotient-persistent structures. Fixed points are transport-stable loci when they preserve standing quotients and invariant overlap observables. Threshold matching is boundary-trace fixation. Dimensional transmutation is role conversion from coupling coordinate to scale anchor. Null endpoints and unsupported selectors do not supply physical standing.

The result does not alter RG practice. It clarifies why RG practice works as a disciplined physical comparison rather than an unconstrained parameter comparison. What is preserved through renormalization is not the numerical immobility of symbols, but standing under lawful scale transport. Formal multiplicity of values, scales, schemes, or effective descriptions does not inherit that standing. Either it remains exploratory bookkeeping, it is licensed by an existing RG continuation locus, or it is retyped as a new target domain with its own admissibility conditions.

A Conceptual Dependencies

The argument draws on the following companion results.

Table 10: Conceptual dependencies.

Source stream	Role in the argument
<i>Gauge-Coupling Parameters and Forced Uniqueness</i>	Raw coupling labels, coupling versus transmutation scale, finite-scale obstruction, fixed-point role, and overlap-level uniqueness.
<i>The Standing Fixity of Physical Constants</i>	Fixed-domain restriction, RG running as lawful transport rather than unlicensed same-domain reassignment, no default measure.
<i>A Classification Theorem for Physical Constants</i>	Five role classes: admissibility-fixed, quotient-dependent, effective, transport-defined, presentation-dependent.

Source stream	Role in the argument
<i>Anchor, Tensor, and Skin</i>	Scheme/presentation distinction, invariant tensor load, anchor/envelope discipline, decomposition of descriptive roles.
<i>A Classification Theorem for Admissible Standard-Model Structures</i>	Boundary/envelope inputs, gauge structure role typing, Standard-Model-level presentation tracking and transport closure.
<i>Boundary-Trace Fixation under Admissibility</i> and continuation-locus materials	Threshold matching as boundary-trace fixation, regime inheritance, and continuation-locus discipline for physical flow.
<i>Continuation-Locus Induction under Admissibility-Bearing Governance</i>	Distinguishes trace succession from standing-bearing continuation; supplies the locus concept used for RG families.
<i>Transport-Closure Locus for Continuous Parameters</i>	Supports the requirement that continuous parameter families close under transport or quotient into invariant structure.
<i>Generation Multiplicity Minimality and Non-Degeneracy</i>	Supplies multiplicity discipline for replicated standing-indistinguishable sectors and unbounded duplication.
<i>Ensemble Embedding Collapse under Standing-Preserving Admissibility</i>	Supplies the formal/inherited/explicit target-domain distinction used to avoid treating formal RG-family multiplicity as inherited standing.
<i>The AMetric Boundary</i>	No default selector for scale, scheme, cutoff, endpoint, representative, or measure before admissibility fixation.
<i>The Structural Inadmissibility of the Null Regime</i>	Null endpoint exclusion: vacuous RG endpoints lack explanatory standing without witness-preserving transport.

B Index of Formal Results

Table 11: Formal results and locations.

Claim	Location
Scale-indexed domains define the comparison object.	Definition 6.1
Overlap witness structure fixes common empirical reconstruction domains.	Definition 6.3
Standing-preserving scale transport requires witness, quotient, and envelope preservation.	Definition 6.4

Claim	Location
Standing-bearing RG content is quotient-level witness structure under scale transport.	Definition 6.5
RG continuation loci identify the standing-bearing target of a formal flow.	Definition 7.1
Formal RG families do not by themselves define transport.	Lemma 7.4
RG transport is not inherited by multiplicity alone.	Proposition 7.5
Cross-domain RG comparisons require target-domain retyping.	Proposition 7.7
Raw coupling values are not the invariant target of RG standing.	Lemma 8.3
RG running is admissible transport, not unsupported reassignment.	Proposition 10.1, Corollary 10.2
Scheme dependence is presentation-dependent when witnesses are preserved.	Lemma 11.2
EFT coefficients are envelope-bound.	Lemma 12.2, Proposition 12.3
Universality classes are quotient-persistent structures.	Theorem 13.2
Fixed points are quotient-stable RG loci.	Definition 14.1, Proposition 14.2
Operator relevance is a transport-role diagnostic.	Proposition 15.2
Threshold matching is boundary-trace fixation.	Proposition 16.2
Matching continuity requires overlap-observable reconstruction.	Proposition 16.3
Dimensional transmutation changes role profile.	Proposition 17.1
There is no default RG selector before admissibility fixation.	Proposition 18.1
Null endpoints lack explanatory standing without transport.	Proposition 19.2
Admissible RG flow is standing-preserving quotient transport.	Theorem 20.1

C Glossary of Role Terms

Table 12: Glossary.

Term	Definition used here
Admissibility	Conditions under which a description can support stable physical reference, comparison, and application.
Standing	Stable empirical and theoretical identifiability across admissible contexts.
Witness ledger	The observables, records, amplitudes, correlations, or empirical identifications used to compare descriptions.
Overlap witness structure	Common reconstruction domain on which observables and empirical identifiers agree after admissible coarse-graining, rescaling, redescription, or matching.
Standing quotient	Equivalence relation identifying presentations that preserve the same physical content.
Envelope	Scale, approximation, cutoff, matching, and boundary conditions licensing an effective description.
Presentation-dependent	Coordinate, scheme, regulator, unit, or basis dependence that does not independently carry standing-bearing load when witnesses are preserved.
Transport-defined	Role whose identity is supplied by lawful transformation between admissible presentations, scales, or regimes.
RG continuation locus	Role-structural locus preserving the same standing-bearing RG comparison target through scale passage.
Formal RG family	Indexed values, scales, schemes, EFTs, cutoffs, or parameter assignments before standing-bearing transport has been supplied.
Explicitly gated target domain	A newly typed comparison domain with its own witnesses, quotient identity, transport maps, envelope, and failure conditions.
Quotient-dependent	Role whose identity is fixed by equivalence-class structure rather than raw representative choice.
Effective	Role bounded by scale, regime, approximation, or EFT envelope.
Anchor	Domain-defining structure required for admissibility, such as law-form, gauge skeleton, or envelope boundary.
Tensor	Quotient-level invariant physical load.
Skin	Presentation surplus with no independent standing-bearing authority.

D Scheme Change and Invariant Extraction

A simple scheme-change calculation illustrates the general point. Let

$$\beta(g) = -b_0 g^3 - b_1 g^5 - b_2 g^7 + O(g^9),$$

and let a coupling redefinition be

$$\tilde{g} = g + a_1 g^3 + a_2 g^5 + O(g^7).$$

The transformed beta function is

$$\tilde{\beta}(\tilde{g}) = \frac{d\tilde{g}}{dg} \beta(g),$$

with the right-hand side re-expanded in $\tilde{g}$. The coefficients beyond the universal sector may change with the scheme. There is no inconsistency. The coordinate description has changed. The physical question is which combinations, predictions, or scaling data remain invariant under admissible comparison.

The calculation supports three role distinctions.

- (S1) The coordinate g is presentation-attached.
- (S2) The flow relation is transport-defined.
- (S3) Invariantly extracted scaling or observable data can carry tensor-level content.

A raw coordinate in this calculation has standing-bearing significance only when supplied with the witness and quotient conditions that make it physical.

E Universality as a Quotient Diagram

The quotient structure of universality may be summarized schematically:

$$\begin{array}{ccc} \text{microscopic presentations} & \xrightarrow{\text{RG}} & \text{long-distance descriptions} \\ \downarrow & & \downarrow \\ \text{presentation differences} & \longrightarrow & \text{standing quotient} \end{array}$$

The upper arrow records coarse-graining and scale transport. The lower row records the fact that many microscopic differences are not part of the long-distance standing target. The right vertical arrow is the quotient operation: it identifies the invariant content preserved as a universality class.

The diagram does not imply that microscopic differences are unreal. It states that those differences lie outside the specific quotient target defined by the universality problem. The same microscopic information may be standing-bearing in a different envelope.

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